10 FISHERIES IMPACT

10.1 Introduction

10.1.1 This section presents the assessment of the fisheries impacts that may arise from the construction and operation of the proposed Hung Shui Kiu New Development Area (HSK NDA) (herein referred to as the "Project"), as per the revised recommended outline development plan (Revised RODP). The Project is located in the northwestern part of the New Territories, midway between the Tuen Mun and Tin Shui Wai (TSW) New Towns (refer to **Figure 2.1**). Details of the key project elements proposed within the Revised RODP (**Figure 2.3**) are described in Chapter 2.

10.1.2 The fisheries impact assessment comprises three major parts: establishment of baseline conditions, evaluation and assessment of fisheries impacts following the criteria and guidelines of Annexes 9 and 17 of the Technical Memorandum on Environmental Impact Assessment (EIAO-TM), and recommendations for mitigation measures where necessary.

10.1.3 Key fisheries issues requiring consideration, as stipulated in **Section 3.4.11** of the EIA Study Brief No. 291/2015, include the following:

- capture fisheries;
- pond / aquaculture; and
- oyster farming.

10.1.4 The fisheries impact assessment includes the following:

- description of the physical environmental background;
- description and quantification of the existing culture fisheries activities;
- description and quantification of the existing fisheries resources (e.g. major fisheries products and stocks);
- identification of parameters (e.g. water quality parameters) and area that are important to fisheries and will be affected;
- identification and quantification of any direct/indirect and on-site/off-site impacts to fisheries, such as permanent resumption and temporary occupation of fish ponds as well as deterioration of water quality of fish ponds, oyster culture areas and their surrounding water bodies;
- evaluation of impacts and making proposals for any practical alternatives or mitigation measures with details on justification, description of scope and programme, feasibility as well as manpower and financial implications including those related to subsequent management and maintenance requirements of the proposals; and
- review the need for monitoring during the construction and operation phrases of the Project and associated works and, if necessary, propose a monitoring and audit programme.

10.2 Environmental Legislation, Policies, Plans, Standards and Criteria

10.2.1 Key guidelines, standards, documents and ordinances/regulations relevant to this fisheries impact assessment are provided below.

- *The Environmental Impact Assessment Ordinance (Cap. 499)*, and its subsidiary legislation – provides guidelines on the environmental impact assessment process.
- *Annex 17 of the EIAO-TM* – sets out the general approach and methodology for assessment of fisheries impacts arising from a project which may affect fishing and aquaculture activities, fisheries resources and habitats as well as *aquaculture* sites. Annex 9 recommends the criteria that can be used for evaluating fisheries impact.
- *Fisheries Protection Ordinance (Cap. 171)* – promotes the conservation of fish and other forms of aquatic life within Hong Kong waters by regulating *fishing* practices to prevent detrimental activities to the fisheries industry.
- *Marine Fish Culture Ordinance (Cap. 353)* – regulates and protects marine fish cultures by designating areas of fish culture zone, granting *licences*, prohibiting unauthorised vessels and any deposition of chemicals or other substances which are likely to cause injury to fish in a fish culture zone.
- *Water Pollution Control Ordinance (Cap. 358)* – aims to control water pollution in waters of Hong Kong. Water Control Zones (WCZs) are designated with individual water quality objectives to promote the conservation and best use of those waters in the public interest.

10.3 Assessment Approach and Methodology

10.3.1 The assessment area, for the purpose of the fisheries assessment, included all areas within the 500 m from the Project area and the areas likely to be affected by the Project and associated works. Ground truthing on the existing conditions of ponds within the assessment area was also undertaken at ponds that were accessible. The assessment area also includes the marine waters of Deep Bay WCZ and North Western WCZ as they are included in the Water Quality Impact Assessment (**Section 5**). The assessment area is shown in **Figure 10.1**.

10.3.2 The existing information regarding the assessment area have been reviewed. Relevant information sources from the private sector and Government include:

- Agriculture, Fisheries and Conservation Department (AFCD) annual reports;
- AFCD Port Survey (2006);
- AFCD websites; and
- Government aerial photos.

10.3.3 Other relevant EIA studies include:

- Deep Bay Link (DBL) EIA (HyD, 2002a);
- Shenzhen Western Corridor (SWC) EIA (HyD, 2002b);
- Upgrading and Expansion of San Wai Sewage Treatment Works and Expansion of Ha Tsuen Pumping Station EIA (DSD, 2002);

- Hang Hau Tsuen Channel at Lau Fau Shan EIA (CEDD, 2009); and
- Expansion of Hong Kong International Airport into a Three-Runway System (AAHK, 2014).

10.3.4 The collated information was evaluated to identify any information gaps relating to the assessment of potential fisheries impacts pertinent to the Revised RODP. In order to obtain the most updated status of existing fishponds within the assessment area, site visits were conducted during January 2016.

10.3.5 AFCD's annual reports, port survey and website provide the most updated information on the development and trends in Hong Kong fisheries. The latest annual fisheries production (2014) is also provided.

10.3.6 The validity of the information compiled during the literature review has been assessed before adoption into this EIA study.

10.3.7 This fisheries impact assessment has been based on the conditions of the currently existing baseline. Impacts were assessed in the absence of mitigation. The construction and operational phase impacts on capture fisheries and mariculture were assessed individually, then cumulatively, in combination with other existing, committed and proposed developments.

10.4 Baseline Conditions

Culture Fisheries

10.4.1 The pond fish / aquaculture industry of Hong Kong is mainly centered in the northwest New Territories. According to AFCD (2015a), local inland fishponds covered approximately 1,140 ha in 2014. These ponds produced 2,001 tonnes of freshwater fish amounting to HK$44 million. The vast majority of these farms engaged in polyculture and farmed species such as Bighead Carp, Grass Carp, Common Carp and Silver Carp in combination with Tilapia or Grey Mullet.

10.4.2 Fishpond farms have been gradually declining in numbers since the 1970s, especially in the Deep Bay area as these farms are being replaced by urban development. **Table 10.1** presents the AFCD figures on fishpond area and annual fish production in Hong Kong from 2002 to 2014.

Table 10.1 Fishpond Area and Annual Fish Production in Hong Kong from 2005 to 2014

Year	Fishpond Area (ha)	Annual Fish Production (Tonnes)
2014	1,140	2,001
2013	1,150	2,187
2012	1,150	2,306
2011	1,130	2,315
2010	1,109	2,190
2009	1,120	2,105
2008	1,160	2,266
2007	1,160	1,930
2006	1,030	1,940
2005	1,030	1,900

Source: AFCD (2005, 2006a, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2015a)

10.4.3 The following baseline information provides a historical overview on the fishponds within the assessment area. During the surveys undertaken for the Upgrading and Expansion of San Wai Sewage Treatment Works and Expansion of Ha Tsuen Pumping Station in 2002, most fishponds were identified as being abandoned. The two active fishponds that were present at the time were lost as a result of the upgrading works undertaken (DSD, 2002).

10.4.4 Under the DBL – EIA Study, two active fishponds with a total area of 1.55 ha were recorded at Tseung Kong Wai and were left undisturbed by the DBL project. Tilapia was the major cultured species, with annual yield of 500 kg (HyD, 2002a). The west side of Ha Tsuen was also surveyed as part of the DBL study. Most of the fishponds were found to be abandoned or inactive, while the only active pond was filled as a result of the DBL project (HyD, 2002a).

10.4.5 Along the coast of Deep Bay, fishponds between Ngau Hom Shek and Ha Pak Nai were surveyed during the SWC EIA Study (HyD, 2002b). Eighteen hectares of fishponds were recorded, of which, 12 ha were active and 6 ha were abandoned. No fishponds were lost as a result of the SWC project.

10.4.6 In a more recent study, four inactive fishponds near Hang Hau Tsuen were recorded (CEDD, 2009).

10.4.7 Based on the review of existing literature, the majority of fishponds within the assessment area are either abandoned, inactive or filled. The only active fishponds were between Ngau Hom Shek and Ha Pak Nai; however, these are records from 2002. The following sections will provide updated results of the fishpond status from recent site visits.

10.4.8 During recent site visits under the current study, a number of different fishpond uses were identified within the assessment area, such uses included recreational fishing, fish pond / aquaculture and inactive fishpond. For the purpose of this study, these various types of fishponds have been separated into two categories, active fishponds and inactive fishponds. Active fishponds are stocked with fish for aquaculture fish farming or for recreational fishing. Inactive ponds are former fishponds that are no longer used or managed by fish farmers and the fish inside the ponds do not yield economic income. Fishing operations including stocking, feeding and harvesting, the presence of aerators and the maintenance conditions of the facilities along the pond bunds were used to determine the status of fishponds (active or inactive). Active fishponds were identified based on evidence of management conditions of the ponds and the presence of fish farming materials and/or equipment and fish farming activities, while inactive fishponds lacked these characteristics.

10.4.9 Nineteen ponds were recorded within the assessment area. Of these, three inactive fishponds were recorded within the Project area where no fishing operations equipment or maintenance conditions were observed. No active fishponds were identified within the Project area. The three inactive fishponds were scattered within the Project area. Of these, two were located between Sai Wai and Fung Kong Tsuen and one was located near San Sang San Tsuen (**Figure 10.2**).

10.4.10 Of the sixteen ponds recorded outside of the Project area, three were active fishponds and thirteen were inactive fishponds. All active fishponds were located along the coastal region between Hang Hau Tsuen and Ngau Hom Sha, at the northern part of the assessment area (refer to **Figure 10.2**). Two active fishponds were located behind locked gates in the northwestern part of the assessment area. The remaining inactive fishponds were scattered along the Deep Bay coast, around Sha Kong Wai, San Sang San Tsuen, Tsing Chuen Wai and Fui Sha Wai.

10.4.11 Refer to **Appendix 10.1** for photographic records of representative fishponds within the assessment area.

Oyster Culture

10.4.12 The culture of oyster has been practiced along the intertidal mud flat of Deep Bay in the northwestern corner of Hong Kong for at least 200 years. Traditionally, oysters are cultured by the bottom culture method with spat collected by laying rock, concrete tile or post as cultch on the mud flat in May or June (AFCD, 2015a). Oysters are grown to marketable size on cultches, which are embedded in the mud by fishermen of which oyster beds are formed. Cultches may be a concrete post, concrete tile, concrete block or stone (AFCD, 2015a). The raft culture method was most commonly adopted, this is where oysters are placed in baskets and suspended from rafts. Production in 2014 was about 121 tonnes (meat only) valued at HK$10 million (AFCD, 2015a).

10.4.13 In Deep Bay, the coastal habitat from Tsim Bei Tsui to Ha Pak Nai consists of 747 ha of mudflats, which sustains 176 ha of oyster beds. Two species of oyster are cultured in the area, *Crassostrea gigas* and *C. rivularis* (HyD, 2002b). During the surveys conducted under the DBL – EIA study, an oyster bed approximately 80 m out from the mangroves at Ngau Hom Shek was found to be actively maintained by local villagers (HyD, 2002a).

10.4.14 As observed on-site during surveys for the current study (from 2011 to 2012 and in 2016), and by reviewing Government aerial photos between 2003 and 2007, there are no oyster rafts currently within the coastal area from Ngau Hom Sha to Lau Fau Shan.

Intertidal Fishing

10.4.15 According to the DBL – EIA Study in 2000, oyster culturing was not the only activity undertaken on the mudflat of Ngau Hom Shek, villagers also fished at these mudflats. Nets were set on wooden poles on the mudflat and traps were connected to the vertical nets to capture fishes, crabs and shrimps. According to the villagers at the time, daily catches yielded to around 30 kg (HyD, 2002a).

Capture Fisheries

10.4.16 In 2014, the Hong Kong capture fishing industry landed approximately 160,789 tonnes of fisheries product valued at $2,530 million (AFCD, 2015b). The industry consisted of about 4,500 fishing vessels and 9,400 fishermen. Fishing activities were mainly conducted in the waters of the adjacent continental shelf in the South China Sea. The majority of the fishing vessels were manned by family members with the assistance of hired crews. Main fishing methods included trawling, long-lining, gill-netting and purse-seining, with the majority of the total catch obtained through trawling. Some recent data on Hong Kong capture fisheries industry is summarised in **Table 10.2**.

Table 10.2 Recent Figures on Hong Kong Capture Fisheries Industry

	Fishing Fleet Size (No. of Vessels)	Local Fishermen Engaged in Capture Fisheries	Production (Thousand Tonnes)	Value of Produce (HK$ Million)
2014	4,500	9,400	161	2,530
2013	4,000	8,800	170	2,238
2012	4,000	8,800	155	2,317
2011	4,000	8,500	171	2,358
2010	3,900	8,200	168	2,100
2009	3,700	7,600	159	2,000
2008	3,800	7,900	158	1,780
2007	4,000	8,500	154	1,530
2006	3,950	8,500	155	1,600
2005	4,150	9,170	162	1,600

Source: AFCD (2005, 2006a, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2015b)

10.4.17 The latest AFCD Port Survey 2006 (AFCD, 2006b) provides the most updated information on capture fisheries in Hong Kong waters, including both fishing operation and fisheries production (adult fish and fry). In general, the highest fish yields in Hong Kong were obtained in the eastern waters (e.g. Tolo Harbour, Crooked Haven, Port Shelter, and Po Toi) and the southwestern waters (e.g. Lamma Island, Cheung Chau and Soko Islands).

10.4.18 The fishing operations within the North Western WCZ and Deep Bay WCZ are dominated by vessels less than 15 m in length. Shrimp trawler, hang trawler, gill netter, purse seiner, sampans and miscellaneous crafts operate near the vicinity of the discharge zones of Urmston Road Outfall and the Tuen Mun River outflow within the North Western WCZ and at the TSW Main Channel outflow within the Deep Bay WCZ, based on the findings from the 2006 Port Survey (refer to **Figure 10.1**). Since 31 December 2012, trawling activities were banned and thus shrimp trawlers and hang trawlers no longer operate within Hong Kong waters. The number of fishing vessels in operation is generally low (10 – 50 vessels) in the Deep Bay WCZ. Within the North Western WCZ, areas with low to moderate numbers of fishing vessels are concentrated around Sha Chau, Lung Kwu Chau, The Brothers and Tai O (100 – 400 vessels), while the remaining waters of this WCZ supported low vessel operations (10 – 50 vessels).

10.4.19 Fishing vessel operations broadly reflected the adult fish productions. In Deep Bay WCZ, adult fish production was low (less than 50 kg/ha). Meanwhile, low to moderate production was found within the North Western WCZ, with production value ranging between 200 – 500 kg/ha at Sha Chau, Lung Kwu Chau and The Brothers. High fisheries production (600 – 1,000 kg/ha) occurred at Tai O.

10.4.20 The overall catch value was low, less than HK$1,000 per hectare annually, within the Deep Bay WCZ. Catch values were relatively higher in the North Western WCZ, with low to moderate catch values (HK$2,000 – HK$10,000 per hectare) near Lung Kwu Chau and The Brothers. Sha Chau and Lung Kwu Chau yielded moderate catch values (HK$5,000 - HK$10,000 per hectare).

10.4.21 All common capture fisheries species can be found within the Deep Bay and North Western WCZs. These species include Scad (Carangidae), Rabbitfish (Siganidae). Croaker (Sciaenidae), Mullet (Mugilidae), Sardine (Clupeidae), Seabream (Sparidae), Anchovy (Engraulidae), Shrimp, Squid and Crab. Shrimp and Croaker were relatively more abundant within the North Western WCZ when compared to other captured species. The abundance of captured species within Deep Bay WCZ were relatively evenly distributed across the WCZ; however, the Scad, Mullet and Seabream distribution tended to be in areas closer to the shore.

10.4.22 No fish fry collection has been recorded within the Deep Bay and North Western WCZs. Fish fry collection is concentrated in the eastern waters of Hong Kong.

10.4.23 Important nursery and spawning grounds for commercial fisheries species were identified at northeastern Lantau (within the North Western WCZ) (AFD, 1998). This area is located approximately 700 m away from the Urmston Road Outfall and 2.4 km away from the Tuen Mun River. The major spawning fish species recorded include Ponyfish (*Leiognathus brevirostris*), Sea Bass (*Lateolabrax japonicas*) and Gizzard Shad (*Clupanodon punctatus*).

10.4.24 In addition to the information available for the Port Survey, fisheries surveys involving fish trawl survey, purse seine survey, gill net survey and hand line survey revealed that the waters of North Western WCZ supported low densities and richness of ichthyoplanton and post-larvae fish. This WCZ also supported mostly fish families of low commercial value (AAHK, 2014)

10.4.25 Effective on 31 December 2012, a ban on trawling to allow the rehabilitation of the damaged seabed and depleted marine resources was administered in Hong Kong waters (AFCD, 2012).

10.5 Identification and Evaluation of Environmental Impacts

Construction Phase

Direct Impact

Direct Loss

10.5.1 Construction works for the Project would occur within the Project area only. The three inactive fishponds identified within the Project area would be lost as a result of the development proposals (refer to **Figure 10.2**). None of these ponds were active (during any of the surveys undertaken from 2011 to 2012 and in 2016) and hence did not yield economic profit. Therefore, no active fishponds would be lost as there are no active fishponds within the Project area.

10.5.2 The affected inactive ponds are small in size, therefore the direct fisheries impact in regards of pond fish / aquaculture is considered negligible to low, when taking into account the potential conversion of inactive fishponds back to active fishpond.

Indirect Impact

Deterioration of Water Quality and Hydrological Condition

10.5.3 Active fishponds are located at least 245 m away from the Project area, while inactive fishponds are located approximately 40 m away from the Project area. Active fishponds are separated from the Project area by built environment. Given the distance and land uses between the ponds and the Project area, indirect impacts (e.g. deterioration of water quality due to run-off from construction works) are unlikely.

10.5.4 Surface run-off during the construction phase could potentially deteriorate the water quality in the Deep Bay WCZ via entering TSW Main Channel and the North Western WCZ via Tuen Mun River. Given the distance between the Project and the North Western WCZ it is unlikely that surface run-off (if it entered Tuen Mun River) would affect the water quality of this WCZ. As for surface run-off into Deep Bay WCZ, according to the Water Quality Impact Assessment, the implementation of mitigation measures as recommended in **Section 5.13** would prevent adverse water quality impacts to the Deep Bay WCZ.

Operational Phase

Indirect Impact

Deterioration of Water Quality

10.5.5 No adverse impacts to fisheries resources are expected to arise from the deterioration of water quality. As mentioned in the Water Quality Impact Assessment (**Section 5**), discharge into the North Western and Deep Bay WCZ would be kept within the discharge limited of the previously approved San Wai Sewage Treatment Works EIA report. As such, it is assumed that the pollution load that would be discharged into these WCZs would be in compliance water quality standards and thus would not significantly deteriorate the water quality of these marine waters.

10.5.6 Surface run-off during the operation phase could potentially deteriorate the water quality in the North Western WCZ and Deep Bay WCZs, however, according to the results from the water quality impact assessment, it is considered that the Project will not cause any net increase in non-point source pollution discharge. With the implementation of recommended stormwater management practices in **Section 5.14**, the water quality impact would be minimised.

10.5.7 In accordance with the EIAO-TM Annex 9 criteria, the assessment of potential fisheries impacts in the absence of mitigation within the assessment area are provided in **Table 10.3** and **Table 10.4**.

Table 10.3 Fisheries Impact to Active and Inactive Fishponds within the Assessment Area

Criteria	Active Fishponds within the Assessment Area	Inactive Fishponds within the Assessment Area
Nature of impact	No active ponds are present within the Project area. No direct / indirect impacts on active fishponds and their pond bunds during construction and operation phases.	Direct impacts on three inactive fishponds during construction phase.
Size of affected area	No active ponds are present within the Project area. No active fishpond would be affected due to the Project.	Three small inactive fishponds.
Loss of fisheries resources / production	No direct loss of fisheries resources / aquaculture production during construction and operation phases.	No direct loss of fisheries resources / production based on current status. Potential loss of fisheries resources / production if the inactive ponds were resumed.
Destruction and disturbance of nursery and spawning grounds	No important nursery and spawning grounds of commercial fisheries species would be disturbed or destroyed during construction and operation phases.	No important nursery and spawning grounds of commercial fisheries species would be disturbed or destroyed during construction and operation phases.
Impact on fishing activity	No active fishing activity would be affected during construction and operation phases.	No loss of fishing activity based on current status. Potential impact on fishing activity if the inactive ponds were resumed during construction and operation phases.
Impact on aquaculture activity	No aquaculturist or aquaculture farm would be affected during construction and operation phases.	No loss of aquaculture activity based on current status. Potential impact on aquaculture activity if the inactive ponds were resumed.
Overall impact before mitigation	**Negligible**	**Negligible** (based on current status) **Low** (if inactive ponds were resumed)

Table 10.4 Impacts to Capture Fisheries in Deep Bay and North Western Water Control Zones (WCZ)

Criteria	Deep Bay WCZ	North Western WCZ
Nature of impact	No impacts are anticipated.	No impacts are anticipated.
Size of affected area	No impacts are anticipated.	No impacts are anticipated.
Loss of fisheries resources / production	No loss in fisheries resources / production.	No loss in fisheries resources / production.
Destruction and disturbance of nursery and spawning grounds	No destruction and disturbance to important nursery and spawning grounds of commercial fisheries species.	No destruction and disturbance to important nursery and spawning grounds of commercial fisheries species.
Impact on fishing activity	No impacts on fishing activity is anticipated as there is no direct or temporary loss of fishing ground within WCZ under this Project.	No impacts on fishing activity is anticipated as there is no direct or temporary loss of fishing ground within WCZ under this Project.
Impact on aquaculture activity	No impacts on aquaculture activity.	No impacts on aquaculture activity.
Overall Impact	**Negligible**	**Negligible**

10.6 Cumulative Impacts from Concurrent Projects

10.6.1 Based on the latest available information, the projects with programmes likely to overlap with the Project construction have been identified in **Section 2**, other Schedule 2 Designated Projects were also identified in **Section 1** (**Table 1.1, Figure 1.2**). Given the nature and locations of the concurrent projects, no cumulative fisheries impact are expected as the concurrent projects would not result in further loss of fishponds.

10.7 Mitigation Measures

10.7.1 The construction and operation phase fisheries impacts of the Project are expected to be negligible. However, the implementation of good site practices during construction and mitigation measures proposed in the Water Quality Impact Assessment (**Section 5**) would help to further minimise impacts to the fisheries resources identified. Mitigation measures are summarised in the following sections.

Construction Phase

Surface Run-off to the Waterbodies

10.7.2 The implementation of water quality mitigation measures as listed below would minimise the potential water quality impacts to ponds:

- The proposed works should preferably be carried out within the dry season where the flow in the storm culvert/river channel/stream is low.

- The use of less or smaller construction plant may be specified in works areas close to the inland water bodies.

- Temporary storage of materials (e.g. equipment, filling materials, chemicals and fuel) and temporary stockpile of construction materials should be located well away from any watercourses during carrying out of the construction works.

- Stockpiling of construction materials and dusty materials should be covered and located away from any watercourses.

- Construction debris and spoil should be covered up and/or disposed of as soon as possible to avoid being washed into the nearby water receivers.

- Construction activities, which generate large amount of wastewater, should be carried out at distance from watercourses, where practicable.

- Mitigation measures to control site run-off from entering the nearby water environment should be implemented to minimise water quality impacts. Surface channels should be provided along the edge of the waterfront within the work sites to intercept the run-off.

- Construction effluent, site run-off and sewage should be properly collected and/or treated.

- Any temporary works site inside the stormwater courses should be temporarily isolated, such as by placing of sandbags or silt curtains with lead edge at bottom and properly supported props to prevent adverse impact on the stormwater quality.

- Proper shoring may need to be erected in order to prevent soil/mud from slipping into the inland water bodies.

10.7.3 With these mitigation measures in place, no adverse water quality impacts are expected, as such there would be no adverse impacts to the fisheries resources at the North Western and Deep Bay WCZs.

Operation Phase

Deterioration of Water Quality of Waterbodies

10.7.4 A surface drainage system will be provided to collect road run-off. The following measures are recommended to ensure road run-off will comply with the standards stipulated in the Technical Memorandum on Standards for Effluents Discharged into Drainage and Sewerage Systems, Inland and Coastal Waters (TM-DSS) for discharges into stormwater drains:

- The road drainage should be directed through silt traps in the gully inlets to remove silt and grit before entering the public stormwater drainage system.

- The silt trap should be regularly cleaned and maintained in good working condition.

10.7.5 Best stormwater management practices and stormwater pollution control plan as discussed in **Section 5.14** would be implemented for the Project to minimise the pollution loading from surface run-off as far as practicable.

10.8 Environmental Acceptability of the Schedule 2 Designated Projects

10.8.1 No unacceptable impacts to the fisheries resources within the assessment area are anticipated from the Schedule 2 Designated Projects identified in **Section 1** (**Table 1.1**). No mitigation measures for fisheries would be necessary for these Schedule 2 projects.

10.9 Evaluation of Residual Environmental Impacts

10.9.1 Residual impacts identified would be the permanent loss of three inactive fishponds within the Project area. As these ponds currently do not contribute to the fisheries production, the impacts are considered to be acceptable.

10.10 Conclusion

10.10.1 Existing fisheries resources within the assessment area include active fishponds (outside of the Project area) and capture fisheries resources of North Western and Deep Bay Water Control Zones (WCZ). The value of capture fisheries in the North Western WCZ is low to moderate while low in Deep Bay WCZ. An important nursery and spawning ground for commercial fisheries species has been identified within the North Western WCZ. No oyster culturing and intertidal fishing were recorded within the coastal area from Ngau Hom Sha to Lau Fau Shan.

10.10.2 Potential fisheries impacts arising from the Project have been assessed. No active fishponds are located within the Project area and therefore would not be lost due to Project. Three inactive fishponds would be lost due to the Project therefore the impact to pond fish culture is considered negligible to low when taking into account the potential conversion of inactive fishponds back to active fishpond.

10.10.3 No unacceptable water quality impacts to the Deep Bay WCZ and North Western WCZ are anticipated from the Project with proper implementation of mitigation measures as recommended in **Section 5.13** and **Section 5.14**. Therefore, monitoring of fisheries resources during the construction and operation phase would not be necessary.

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